Comparative Evaluation of Methods for Position-Independent Voiced-Consonant Detection in Isolated Words

Bunyod Suvonov and Giorgio Aman Soetomo
Student IDs: 522370990020 and 523370990002

Abstract—This paper presents a comparative evaluation of methods for position-independent full-word detection of four voiced consonants, d, n, r, and v, using recordings from Speech Commands v0.02. The target consonant may appear at the beginning, middle, or end of the word, so the detection problem is formulated as letter presence over the whole active word. A 23-keyword benchmark is constructed with six positive keyword types for d, n, and r, and all four available positive keyword types for v. We compare five classical detectors and one multilayer perceptron (MLP) baseline: global fast Fourier transform (FFT) template matching, Mel-frequency cepstral coefficient (MFCC) mean-template scanning, MFCC + dynamic time warping (DTW), local short-time Fourier transform (STFT) mean-template scanning, local STFT k-nearest-neighbor (k-NN) scanning, and a small MLP operating on full-word log-Mel features. The FFT/STFT spectrum detectors are implemented in MATLAB, while the MFCC/DTW and MLP detectors provide comparison methods from speech recognition and neural-network modeling. On speaker-independent test sets balanced separately for each target consonant, the classical methods achieve macro F1 between 66.30% and 67.16%, while the MLP reaches 85.87% macro F1 and 85.80% macro accuracy. In these experiments, coarse template detectors have high recall but poor precision under word-level labels and weak phoneme alignment, whereas the trained MLP gives a sharper full-word decision boundary.

I. Introduction

Speech is a time-domain waveform, but many phonetic cues become much clearer after short-time spectral analysis. Sampling, Fourier analysis, and local spectra make physically meaningful structure visible. This work uses those tools as part of a controlled comparison of detectors for deciding whether selected voiced consonants are present in spoken isolated words. Speech Commands v0.02 is a public corpus of short, single-word speech recordings designed for limited-vocabulary keyword-spotting research; here it is used as a source of isolated-word utterances with word labels.

The critical methodological issue is position. A target consonant can occur in different parts of an isolated word, for example d in dog and bed, or r in right and zero. Therefore, we study position-independent detection of four voiced consonants:

- d: voiced alveolar stop,
- n: voiced nasal stop,
- r: voiced liquid,
- v: voiced labiodental fricative.

The comparative study is organized around three questions:

1) How can local spectral templates be extracted for consonants in word-level recordings?
2) How well do classical spectrum, Mel-frequency cepstral coefficient (MFCC), and dynamic time warping (DTW) detectors work when the target consonant may occur anywhere in the word?
3) How does a multilayer perceptron (MLP) classifier trained directly on whole-word features compare with the classical detectors?

The contribution is a controlled comparison of six detectors on the same 23-keyword Speech Commands benchmark, with identical letter labels, speaker-independent splits, and metrics. Local target templates are constructed from approximate phoneme-position priors, and each local detector scans the entire active part of the word at test time.

II. Related Work

Classical speech recognition relied heavily on short-time spectral analysis and template matching. Rabiner and Schafer describe the central role of short-time processing in speech analysis [1]. Davis and Mermelstein introduced MFCCs as a compact and effective spectral representation for speech recognition [2]. Sakoe and Chiba proposed DTW to align speech templates under speaking-rate variability [3]. These ideas motivate the MFCC and DTW baselines used here.

Modern speech systems increasingly use learned models rather than direct template matching. Hinton et al. summarized the impact of deep neural networks on acoustic modeling [4]. For limited-vocabulary speech tasks, Sainath and Parada showed that compact neural networks can outperform conventional baselines in keyword spotting [5]. Speech Commands was introduced by Warden as a public benchmark for limited-vocabulary keyword spotting and has been widely used to evaluate isolated-word speech recognition methods [6].

Our comparison therefore includes MATLAB spectrum methods based on the fast Fourier transform (FFT) and short-time Fourier transform (STFT), MFCC/DTW baselines, and an MLP trained on the same Speech Commands recordings.

III. Problem Formulation and Theory

A. Sliding-Window Letter Detection

Let $x[n]$ be a word recording and let $[t_s, t_e]$ denote its active speech region after simple energy-based voice activity detection. We slide a fixed analysis window across that active region and obtain local segments

$$w_m[n] = x[n + t_m], \quad m = 1, \ldots, M. \tag{1}$$

For a target consonant c, a template bank $B_c = \{b_{c,1}, \ldots, b_{c,K_c}\}$ is built from training recordings. A generic scan detector uses

$$S_c(x) = \max_m g_c(\phi(w_m), B_c), \tag{2}$$

where $\phi(\cdot)$ is the chosen local feature map and g_c is the method-dependent bank score: maximum cosine similarity, negative DTW distance, or top-k average cosine similarity. The final decision is

$$\hat{y}_c(x) = \begin{cases} 1, & S_c(x) \geq \tau_c, \\ 0, & S_c(x) < \tau_c, \end{cases} \tag{3}$$

with threshold τ_c chosen on the validation set. The output is therefore a target-specific score $S_c(x)$ and binary decision $\hat{y}_c(x)$ for each consonant $c \in \{d, n, r, v\}$. The task is not a four-class choice among the four consonants; it is four separate presence-detection problems, each asking whether one target consonant occurs somewhere in the word.

B. Approximate Phoneme-Position Prior

Speech Commands provides word labels but not phoneme boundaries. To initialize target templates, we map each selected keyword to a simple ARPABET (ARPA phonetic alphabet) pronunciation and approximate the target-phoneme center by a uniform partition of the active word. If a word has L phonemes and the target occurs at zero-based phoneme index i, we estimate its center as

$$\hat{t}_i = t_s + \frac{i + 0.5}{L}(t_e - t_s). \tag{4}$$

This weak alignment assumption is used only during training to choose template windows. Word labels are also used to define positive and negative examples and to compute ground-truth evaluation metrics. They are not used as inputs to the detector at test time: the detector receives only the waveform-derived spectral features and then scans the whole active region.

C. Short-Time Spectrum, MFCC, and DTW

For a frame window $h[n]$, the short-time Fourier transform (STFT) is

$$X_{w,r}[k] = \sum_{n=0}^{N-1} w[n + rH]h[n]e^{-j2\pi kn/N}. \tag{5}$$

For each scanned 100 ms analysis window, the local STFT feature is constructed by computing the log-magnitude STFT inside that window and then averaging over its short-time frames:

$$\phi_{\text{STFT}}(w)[k] = \frac{1}{M_w} \sum_{r=1}^{M_w} \log\left(|X_{w,r}[k]| + \epsilon\right). \tag{6}$$

Thus $\phi_{\text{STFT}}(w)$ is a single spectrum vector describing the local frequency content of one consonant-sized region of the word. The local STFT detectors compare this vector with target-consonant templates using cosine similarity. This keeps the detector directly interpretable in the frequency domain.

Mel-frequency cepstral coefficients (MFCCs) compress the short-time spectrum into a perceptually motivated cepstral representation. If $E_m[b]$ is the Mel-band energy of frame m and band b, then

$$c_m[q] = \sum_{b=0}^{B-1} \log E_m[b] \cos\left(\frac{\pi(b + 1/2)q}{B}\right). \tag{7}$$

Dynamic time warping (DTW) then compares two MFCC sequences by the minimum dynamic-programming path cost under local step constraints [3]. In our detector, DTW is applied between every scanned test window and a small bank of representative target templates.

D. MLP Detector

For the multilayer perceptron (MLP) method, each full word is converted into a resized log-Mel spectrogram vector $z(x)$. We train one binary MLP per target consonant:

$$p_c(x) = \sigma(W_2 \rho(W_1 \tilde{z}(x) + b_1) + b_2), \tag{8}$$

where $\tilde{z}(x)$ is standardized, $\rho(\cdot)$ is the rectified linear unit (ReLU), and $\sigma(\cdot)$ is the sigmoid function. This model sees the entire word at once and is not limited to one local template.

IV. Dataset and Experimental Setup

A. Dataset and Target Selection

We use the full Speech Commands v0.02 dataset [6]. From it, we select 23 keywords: backward, bed, bird, cat, dog, down, five, forward, four, go, house, marvin, nine, no, on, right, seven, three, up, visual, wow, yes, and zero. Six of these keywords, cat, go, house, up, wow, and yes, do not contain any of the four target consonants and are included as target-free negative examples. To keep the computation comparable across all methods, each keyword is capped at 150 training recordings, 60 validation recordings, and 120 test recordings, giving 7,590 recordings in total.

The split is speaker-independent. In Speech Commands filenames such as 00b01445_nohash_0.wav, the token before _nohash_ is a hexadecimal speaker identifier. We convert this token to an integer and then take modulo 100, which maps each speaker to a deterministic bucket from 0 to 99. Buckets 0–19 are assigned to the test set, buckets 20–29 to the validation set, and buckets 30–99 to the training set. Thus the split is approximately

70%/10%/20% train/validation/test by speaker before the per-keyword caps are applied. This is different from a random 80%/20% utterance split: every recording from the same speaker goes to the same partition, which avoids evaluating on the same speaker identities seen during training.

Table I shows the positive keywords for each target consonant. The labels I/M/F denote initial, medial, and final target positions in the approximate pronunciation. For lexical comparability, the d, n, and r detectors use six positive keyword types, which matches the number of usable d keywords in the selected corpus. The v detector uses all four selected Speech Commands keywords containing /v/, because no additional clean /v/ keyword folders are available in this corpus. Some words contain more than one target consonant, so they can be positive examples for more than one binary detector.

Table I lists target-keyword assignments rather than distinct dataset keywords. It contains 22 positive assignments because some keywords are positive for multiple targets; for example, marvin is positive for n, r, and v. After removing such overlaps, the table contains 17 distinct positive keywords. Together with the six target-free negative keywords, this gives 23 selected keywords in the benchmark.

For each target consonant, the word label is converted into a binary presence label using the pronunciation table. For example, marvin is positive for n, r, and v, but negative for d. The negative class is sampled from the remaining selected keywords that do not contain that target. The negative examples are subsampled to match the number of positive examples, so every detection problem is balanced. During testing, these labels are used only to score the predictions, not to generate them.

B. Assumptions and Preprocessing

The experiments use the following assumptions:

- Speech Commands provides word labels but not phoneme boundaries.
- Active speech can be isolated by an energy threshold on 20 ms frames with 10 ms hop.
- Approximate phoneme timing can be estimated by dividing the active region according to pronunciation length.
- Detection is position-independent at test time, so all local methods scan the full active word.

Each recording is mean-removed and peak-normalized. The active region is then detected by thresholding short-time energy at 10% of the utterance maximum. Local detectors use a 100 ms analysis window, a 25 ms STFT frame, 10 ms hop, and 512-point FFT. MFCC features use 26 Mel filters and 13 cepstral coefficients.

C. Implemented Methods

We compare six methods. The global FFT, local STFT mean-template, and local STFT k-nearest-neighbor (k-NN) detectors form the MATLAB spectrum-based implementation. MFCC mean-template and MFCC + DTW are speech-recognition baselines, and the MLP is trained for the same detection task. The MLP is included as a learned whole-word baseline, not as an a priori preferred method; all methods use the same train/validation/test protocol. Table II summarizes the models, assumptions, and decision rules.

D. Algorithm Structure

Figure 1 connects the method names to the algorithmic blocks. A method is a concrete feature and scoring choice, while an algorithm block is a reusable procedure. The six methods are grouped into three branches. Branch A contains the one whole-word classical method, Branch B contains the four local-template classical methods, and Branch C contains the one learned MLP method. Algorithms 2 and 3 are the training and inference stages for the classical branches: Algorithm 2 builds the target templates during training, and Algorithm 3 scores a test recording.

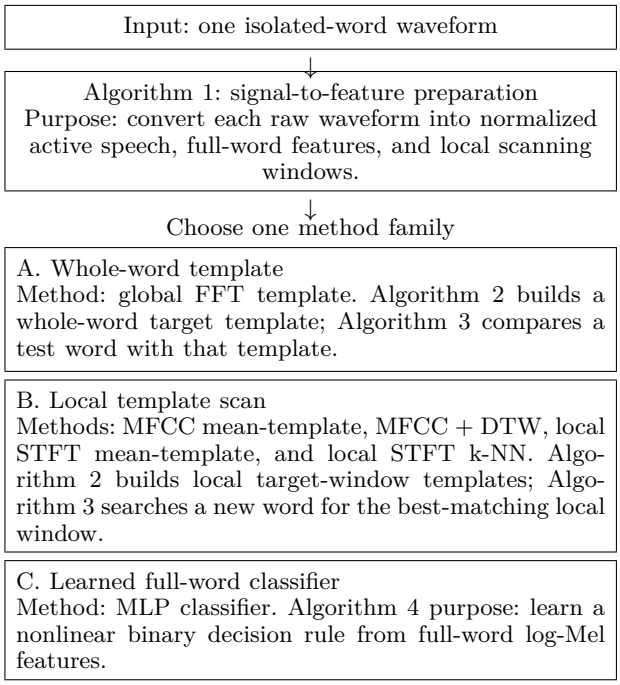

Fig. 1: Block diagram linking the six implemented methods to the four algorithm structures. Branches A and B share the classical train-template/test-score stages but use different feature scopes and scores.

The numbering separates reusable stages, not one algorithm per method. Algorithm 1 is common preprocessing: waveform normalization, active-speech detection, and generation of both full-word features and local scanning windows. This block is shared by all six methods.

The four algorithm types have different purposes. Algorithm 1 is a signal-to-feature preparation algorithm.

TABLE I: Target consonants, phonetic roles, and positive keyword assignments used in the experiments

Target	Phonetic role	Positive keywords used for detection
d	voiced alveolar stop	backward(F), bed(F), bird(F), dog(I), down(I), forward(F)
n	voiced nasal stop	down(F), marvin(F), nine(I/F), no(I), on(F), seven(F)
r	voiced liquid	forward(M), four(F), marvin(M), right(I), three(M), zero(M)
v	voiced labiodental fricative	five(F), marvin(M), seven(M), visual(I)

TABLE II: Implemented methods, feature spaces, assumptions, and decision rules

Method	Feature / model	Training assumptions and structure	Test-time decision rule
Global FFT template	Whole active-word log FFT spectrum	Build one positive mean template per letter from whole-word spectra; threshold tuned on validation set	Detect the letter if cosine similarity to the positive template exceeds the tuned threshold
MFCC mean-template scan	13-dimensional MFCC mean of each 100 ms local window	Use pronunciation-based target windows to form initial/medial/final positive template means	Scan the active word and take the maximum cosine similarity over all windows and templates
MFCC + DTW	Sequence of MFCC vectors inside each 100 ms local window	Use pronunciation-based target windows to form a representative positive bank; tune template count and Sakoe–Chiba band on validation data	Scan the active word and use the negative minimum DTW distance as the score
Local STFT mean-template scan	Average local log spectrum of each 100 ms window	Use pronunciation-based target windows to form initial/medial/final positive spectral templates	Scan the active word and take the maximum cosine similarity over all windows and templates
Local STFT k-NN scan	Bank of local average log spectra	Store representative target windows; tune k on validation data	Scan the active word and use the best top-k average cosine similarity as the score
MLP classifier	Whole-word resized log-Mel spectrogram vector	Train one binary MLP per letter with ReLU, sigmoid output, Adam, and early stopping	Use the output probability and a validation-tuned threshold to decide letter presence

Algorithm 2 is a training-time template-learning algorithm for classical detectors. Algorithm 3 is a test-time search-and-scoring algorithm for classical detectors. Algorithm 4 is a learned classification algorithm that trains and applies a binary neural network.

Algorithm 2 builds classical templates. For the global FFT method, the positive training recordings are represented by whole-word log spectra and averaged to form one target template. For the local methods, positive words are first mapped to their pronunciation strings, the approximate target position is estimated using (4), and the closest 100 ms local window is extracted. The selected local windows form either mean templates, representative DTW templates, or a k-NN template bank.

Algorithm 3 scores classical test recordings. The global FFT method computes one cosine similarity score between the test word and the whole-word target template. The local-template methods scan all active-word windows and keep the best target score over time. The score is maximum cosine similarity for mean-template matching, negative minimum DTW distance for MFCC + DTW, and best top-k average cosine similarity for local STFT k-NN. In all cases, the final threshold is selected on the validation set.

Algorithm 4 is the MLP branch. It converts the active word into a resized full-word log-Mel vector, standardizes it using training-set statistics, trains one binary MLP per target consonant, and uses a validation-tuned probability threshold for the final decision.

V. Spectral Analysis

Figure 2 shows the average local spectra extracted from the estimated target windows. This figure is descriptive rather than a complete classifier: because all four targets are voiced consonants, their averaged spectra share low-frequency voicing structure and are not expected to be visually separated by a single curve. The methods distinguish consonant presence by computing a target-specific score over whole-word or local-window features. For the classical methods, this score is based on cosine similarity, DTW distance, or k-NN similarity to target templates; for the MLP, it is a learned probability from full-word log-Mel features. The stop consonant d shows a short transition-like spectrum, n concentrates more low-frequency energy because of nasal resonance, r has a smoother vowel-like transition, and v contributes frication energy from a voiced labiodental constriction.

Figure 3 validates the task definition by showing highlighted windows at different target positions: d is captured at the end of bed, n at the end of seven, r in the middle of zero, and v at the beginning of visual. These examples illustrate the full-word scanning problem.

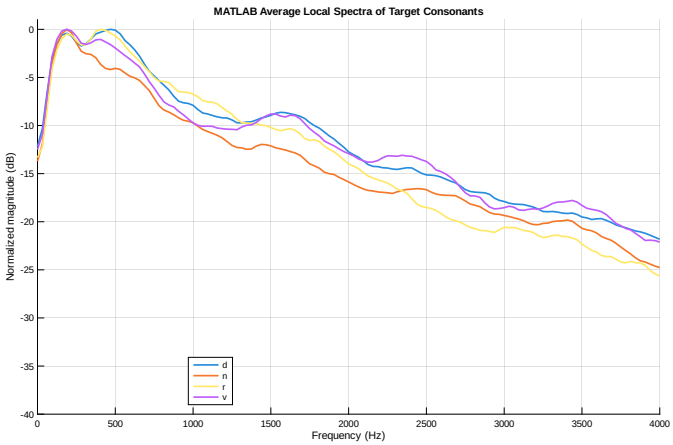

Fig. 2: MATLAB average local spectra of the four voiced consonants extracted from approximate target windows.

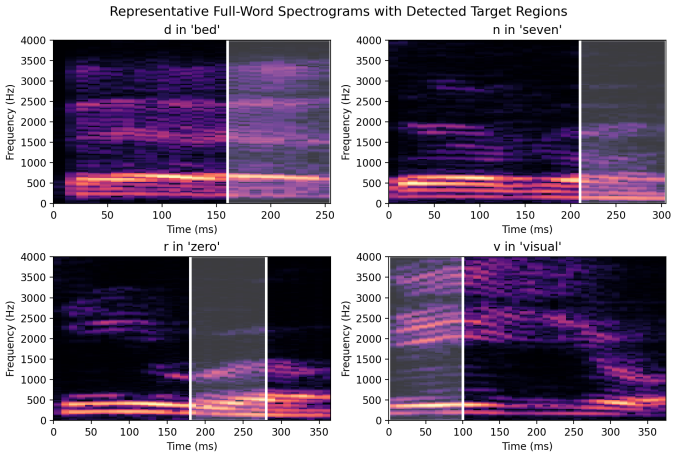

Fig. 3: Representative full-word spectrograms with highlighted target windows. The examples include initial, final, and medial consonant positions.

VI. Results and Comparisons

A. Overall Comparison

Table III reports the average precision, recall, F1, and accuracy over the four balanced binary detection tasks. The average metrics show that the classical detectors are recall-heavy but imprecise, while the MLP gives more balanced precision and recall.

The MATLAB spectrum implementation was also run separately for the FFT/STFT detectors. It gave comparable macro F1 values: 66.75% for global FFT, 65.96% for local STFT mean-template scanning, and 66.49% for local STFT k-NN scanning. These values agree with Table III: classical spectrum scanning is useful for spectral interpretation, but in this benchmark it is less selective than the MLP under weak phoneme alignment.

Figure 4 presents the same comparison graphically. The five classical methods cluster closely together because they all depend on weakly aligned target templates. Once

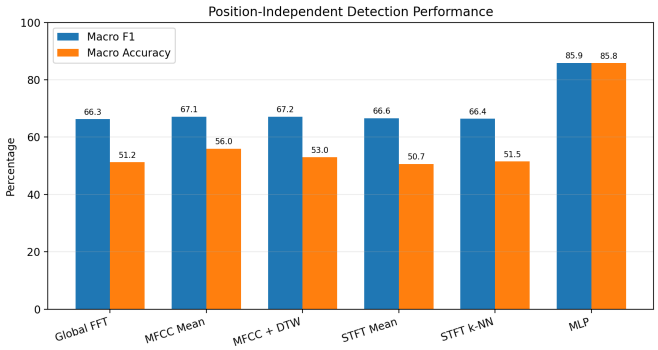

Fig. 4: Macro F1 and macro accuracy for all implemented methods. Percentage labels are printed above the bars to make close classical baseline values readable.

Fig. 5: Per-letter detection F1 for an MFCC/DTW baseline, a local-spectrum baseline, and the MLP classifier. Percentage labels are printed above the bars.

the target is allowed to occur anywhere in the word, coarse positive-template scanning becomes difficult to make selective.

B. Per-Letter Comparison

Figure 5 compares one MFCC/DTW baseline, one local-spectrum classical detector (STFT k-NN), and the MLP on each letter. In the test results, the MLP obtains the highest F1 for all four consonants: 83.83% for d, 84.06% for n, 89.20% for r, and 86.39% for v.

Among the classical methods, performance is especially limited for letters whose target windows are hard to isolate with only word-level supervision. This is expected: the classical detectors know how the target should look locally, but they do not learn a global representation of the entire word.

C. Discussion

The experiments lead to three main conclusions.

1) Weak phoneme alignment is the main bottleneck for classical detectors. The classical methods achieve recall around 90–98%, but precision only around 50–55%. They often find a spectrally similar local region somewhere

TABLE III: Average detection performance across the four target consonants on the speaker-independent test split

Method	Macro precision (%)	Macro recall (%)	Macro F1 (%)	Macro accuracy (%)
Global FFT template	50.74	96.04	66.30	51.22
MFCC mean-template scan	54.66	89.81	67.14	55.98
MFCC + DTW	51.65	96.15	67.16	52.95
Local STFT mean-template scan	50.36	98.47	66.62	50.66
Local STFT k-NN scan	50.86	95.85	66.40	51.49
MLP classifier	85.50	86.27	85.87	85.80

in the word, even when the target consonant is absent. This high-recall/low-precision behavior is consistent with positive-template scanning under approximate target localization.

2) MFCC/DTW baselines and spectrum baselines are both informative. MFCC mean-template and MFCC + DTW represent standard speech-recognition feature and alignment methods. Global FFT, local STFT mean-template, and local STFT k-NN are MATLAB spectrum methods based on direct Fourier analysis. In the full-word setting, none of the classical methods dominates decisively; all are constrained by the same lack of exact phoneme boundaries.

3) The MLP gives the highest scores in this study. The MLP sees the entire log-Mel representation of the word and learns nonlinear boundaries between positive and negative examples. In the measured results, its macro F1 is 85.87%, compared with roughly 66–67% for the classical methods, and its precision and recall are more balanced.

There are still limitations. The target timing is approximate rather than forced-aligned, so the local template banks may include neighboring vowel or transition energy. The expanded design uses six positive keyword types for d, n, and r, but the v detector remains limited to four positive keyword types by the available Speech Commands vocabulary. Also, the selected d keywords contain initial and final /d/ but no medial /d/ example. A phoneme-aligned corpus with broader vocabulary would allow a cleaner study of purely acoustic consonant spotting and stronger control over target position.

VII. Conclusion

This work analyzed the spectra of four voiced consonants in MATLAB, implemented MATLAB spectrum-based detectors, evaluated MFCC/DTW baselines, and trained an MLP on Speech Commands recordings. The task is formulated as position-independent letter detection inside words.

Under that formulation, the five classical detectors all show similar high-recall/low-precision behavior, while the MLP gives the highest measured result with 85.87% macro F1 and 85.80% macro accuracy. Overall, Fourier-based spectrum tools remain useful for spectral interpretation and baseline detection, but once target timing becomes uncertain, learned whole-word representations can be more robust than weakly aligned local templates.

References

[1] L. R. Rabiner and R. W. Schafer, Digital Processing of Speech Signals. Englewood Cliffs, NJ: Prentice-Hall, 1978.

[2] S. B. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 28, no. 4, pp. 357–366, 1980.

[3] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. 26, no. 1, pp. 43–49, 1978.

[4] G. Hinton, L. Deng, D. Yu, G. E. Dahl, A.-r. Mohamed, N. Jaitly, A. Senior, V. Vanhoucke, P. Nguyen, T. N. Sainath, and B. Kingsbury, "Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups," IEEE Signal Processing Magazine, vol. 29, no. 6, pp. 82–97, 2012.

[5] T. N. Sainath and C. Parada, "Convolutional neural networks for small-footprint keyword spotting," in Proceedings of Interspeech, 2015, pp. 1478–1482.

[6] P. Warden, "Speech commands: A dataset for limited-vocabulary speech recognition," arXiv preprint arXiv:1804.03209, 2018.